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Blockchain-based Proxy Re-Encryption Scheme for Data sharing in Internet of Vehicles

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***Abstract*—With the rapid development of the Internet of Vehicles(IoV), the value of vehicle data has received more and more attention and it cannot be fully utilized without data circulation and sharing. However, due to problems such as data security, and low quality, the sharing of vehicular data grows slowly. Therefore, this paper proposes a user-centric vehicular data sharing scheme based on consortium blockchain and proxy re-encryption to solve the problem of data silos among current vehicles, vehicular service providers, IoT devices, and other beneficiaries. Besides, we introduce a trusted decentralized identifiers mechanism based on IoV to solve the authentication problem of the IoV entities. Several experiments and security analysis show that the scheme can achieve trusted identity authentication in data sharing of IoV, and effectively protect the user's data privacy, while at the same time reaching the actual business scenario-level time delay.**

***Index Terms*—blockchain, data sharing, data privacy, IoV, proxy re-encryption**

I. Introduction

In the coming 5G era, the high-throughput and low-latency IoV will be marketed in large areas, and the rapid development of IoV represents the main trend of intelligent transportation[1]. IoV is not just connecting vehicles to vehicles, it is also connecting vehicles to pedestrians, vehicles to infrastructure (signal lights, etc.), vehicles to networks, vehicles to clouds. It plays an increasingly important role in improving the efficiency of transportation systems. At the same time, IoV will establish a new business model of vehicular service and promote the development of autonomous driving technology, and provide drivers with more intelligent and efficient driving services. A lot of vehicular data are saved by vehicle devices in the form of electronic records during the driving process. With the development of autonomous driving technology, big data, cloud computing, and other technologies, vehicular data has gradually become the focus of vehicle manufacturers and related companies in the automotive sector. For example, different users can collect driving data such as driving time, location, acceleration and deceleration, and turning through various onboard sensor devices and choose to share it to the related platforms for analysis and modeling, which is used to depict the user's driving risk level and serve as a reference for insurance pricing. Data sharing among vehicles or vehicles and other roles is conducive to improving the user's driving experience and the quality of intelligent transportation. Vehicular data has become an indispensable resource for the IoV application, and the sharing of vehicular data has become the consensus of the entire vehicular industry. However, due to the security and efficiency of data sharing, the sharing of vehicular data became extremely difficult at first. Many users can not share their data among different vehicles and data silos are formed. Moreover, the accumulation of vehicular data consumes a lot of resources and the value of vehicular data itself is not reflected.

Therefore, many vehicular data sharing solutions are trying to ensure the security of vehicular data while achieving high efficiency so that it can be shared among different vehicles to make full use of the value of vehicular data. In [2], the authors use the cloud-based data sharing solution to exchange vehicle information in IoV and introduce third-party auditors in the cloud to verify the integrity of shared data to improve the efficiency of audit tasks. A new mobile P2P architecture in the IoV is proposed in[3], which can make each vehicle node under the same driving test unit to share resources, reducing the bandwidth pressure on the server-side. These methods have a certain role in promoting the data sharing of the traditional IoV architecture, but the security issues in distributed vehicular network data sharing are still unresolved.

Blockchain is a decentralized distributed ledger technology that was applied in bitcoin [4]. In recent years, the blockchain has developed rapidly, and people have begun to apply it to the sharing of vehicular data [5]. Blockchain has non-tamperable and good cryptographic features to ensure that the data can be traced back. The smart contract of blockchain can be used as a pre-agreed program that can automatically execute once it is triggered[6].Setting blockchain as the underlying infrastructure, each subject participating in vehicular data sharing acts as a blockchain node, and the data sharing operations between the

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nodes can be used as a record on the public ledger, which can be traced and reviewed and greatly facilitates accountability afterward. There have been many attempts[7]-[10] to share vehicular data sharing schemes based on Blockchain, but many of them are limited to the conceptual level. Because of the use of blockchain technology in IoV, the performance of the solution must be considered first, such as latency and throughput, etc. Second, it depends on whether it helps improve data security. Therefore, there are many issues to be considered in actual blockchain-based IoV data sharing applications.

In this paper, we propose a user-centric vehicular data privacy sharing scheme based on consortium blockchain [11]-[17] and proxy re-encryption [18], which is compatible with existing cloud-based data storage and combines the trusted decentralized identifiers mechanism (DID)[19] to solve data silos issues among current vehicles, vehicular service providers, IoT vehicular devices, etc. And we give the rights of controlling vehicular data sharing back to individual users. Our contributions can be generalized as follows:

- We combine DID and consortium blockchain to ensure the identities of the entities involved in IoV data sharing, which aims at solving the problem of trusted identity verification.
- We propose a user-centric vehicular data privacy sharing scheme using proxy re-encryption to improve the level of data security in the process of vehicular data sharing, and also meet the performance requirements of actual application scenarios.
- We verified the feasibility of our proposed solution through some experiments and perform a security analysis of it.

The remainder of the paper is organized as follows: We systematically summarized the work related to vehicular data sharing in section II. In section III, we introduce some background concepts. In section IV, we present our overall architecture and the workflow for addressing the current challenges of vehicular data sharing will be introduced. Experimental details and results of our proposed scheme will be described in section V. A theoretical security and performance analysis is proposed in section VI. Finally, we conclude the paper in section VII.

II. Related Work

The predecessors of data sharing in IoV have done a lot of work. Most of them mainly focus on two aspects, one is the performance of the data sharing system, and the other is the data security problem in the process of data sharing.

In [2], Runa used the cloud-based method to exchange vehicle information in IoV, vehicles can join different interest groups to form a social network where they can facilitate the modification or sharing of data in the cloud. Kong Q proposed the improved Parlier cryptosystem to achieve sensory data aggregation and preserve location privacy, thereby enabling data sharing in IoV[20]. Garg Sahil proposed "Sec-IoV", a multi-stage model for anomaly detection, to ensure vehicle-to-vehicle (V2V) communication in IoV settings, to explore solutions to address the security risks of highly dynamic deformation and distributed characteristics of the vehicle network[21]. Fan proposed a two-way authentication protocol to protect the driver's privacy, using transparent computing in the protocol and client hardware to improve the quality of the user experience[22].

In [23], the authors proposed a quality-driven data sharing mechanism based on the consortium blockchain, which ensures trust in on-chain and off-chain data. [24] proposed a credibility framework called Vcash for identifying denial of traffic services to solve the problem of trust in the application level. Each vehicle directly communicates with RSU for traffic incident propagation and verification, and shares verified traffic data in the Vcash system. In [25], the authors developed a lightweight vehicle-mounted blockchain security (LVBS) data sharing framework in drone-assisted IoV and developed a credit-based consensus algorithm in the blockchain to track the behavior of the nodes and record transactions data to improve the efficiency and security of reaching consensus. In [26], the authors proposed a novel blockchain-based architecture named BlockAPP for vehicle authentication, privacy protection, and seamless access control to IoV.

Inspired by NuCypher's proxy re-encryption solution[27], we try to use encrypted access control performed through a decentralized network to move beyond the limitation of using a consensus network to safely store and process private encrypted data. The solution supports and provides a secure infrastructure for decentralized applications to share sensitive data varying from vehicular data sharing to identity management.

III. Preliminaries

In this section, some preliminaries used in our blockchain-based data sharing scheme are introduced.

A. *Consortium Blockchain*

Consortium blockchain refers to a blockchain with several organizations in management and it only allows the institutions within the system to read, write data, and send transactions. Due to the limited nodes, the consortium blockchain has the advantages of faster transaction speed, stronger access control, lower transaction cost, etc. It is easier to seek common ground while reserving differences in terms of interests. [28] proposed a consortium blockchain framework to solve the problem of detecting malicious code in malware and extracting corresponding evidence in mobile devices. Compared with the previously released algorithm, the new method can achieve faster detection speed for the reason that the consortium blockchain has better scalability and higher throughput. Therefore, the consortium blockchain is more in line with enterprise-level requirements in IoV and the application is more targeted and efficient. At the same time, the consortium blockchain can also maximize the data management characteristics of the blockchain technology in the enterprise application market, reducing the trust cost of transactions in different units' non-trust environments in data sharing scenarios.

B. Distributed Identity(DID)

In the real world, there are a variety of data used to describe the identity of entities and relationships between entities, such as ID cards, driving licenses, etc. Distributed identity(DID) is a distributed identity protocol [29], which can be implemented based on the existing blockchain. The real entity (person or object) can obtain a unique identity on the blockchain through DID. Through DID, a credible electronic credential can be generated for users, which can be verified by the institution after authorization by users, thereby simplifying business processes and reducing the risk of private data leakage. DID can establish a trusted identity and it is widely used in vehicular data sharing affairs.

C. Proxy Re-encryption

Proxy re-encryption is to entrust a trusted third party or a semi-honest proxy to convert the ciphertext encrypted by the public key of one party into a ciphertext that can be unlocked by the another party's private key to implementing key sharing[30]. In the current data sharing scenario, the users' data stored in the cloud is in encrypted form. Since the data owner does not fully trust the cloud service provider, the key to decrypt the ciphertext cannot be sent to the cloud. And proxy re-encryption can realize cloud ciphertext data sharing without leaking the decryption key of the data owner.

In the classic proxy re-encryption scenario, there are several roles. Alice(sk_A, pk_A) is the data owner and she wants to transfer data M to Bob(sk_B, pk_B) with a semi-honest proxy (such as the company providing cloud services) between them. The specific communication process is:

Alice: Encrypt the plaintext M with her public key. The M is the content that A wants to send to B.

$$C_{pkA}=\text{encrypt}(pk_A,M) \tag{1}$$

Alice: Send the ciphertext C_{pkA} generated by equation (1) to the semi-honest Proxy and generate the conversion key $rk_{A\text{->}B}$ for it. This key is generated by A for the Proxy.

$$rk_{A\text{->}B}=\text{rekey}(sk_A,pk_B) \tag{2}$$

Proxy: The re-encryption key generated by A ($rk_{A\text{->}B}$) in equation (2) converts the ciphertext C_{pkA} into the ciphertext that can be decrypted by Bob's private key. The Proxy only provides the calculation conversion service and it can't obtain the plaintext and sends the generated ciphertext C_B to Bob.

$$C_B = \text{reencrypt}(rk_{A\text{->}B}, C_{pkA}) \tag{3}$$

Bob: Decrypted C_B generated by equation (3) with sk_B to get the plaintext M that A wants to secretly share.

The above is the whole process of proxy re-encryption, and it can also perform multiple encryptions according to the number of semi-entities to achieve data security.

IV. PROPOSED SCHEME

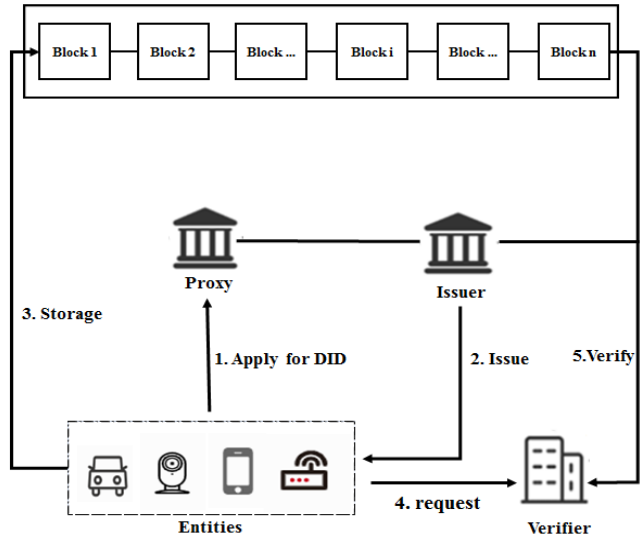

Fig. 1. Trusted identity mechanism

A. Trusted identity authentication

We use the DID mechanism to ensure the trustworthiness of the vehicular data sharing participants including vehicles, IoT equipment, vehicular service providers, mobile devices, etc. It can minimize some malicious attacks such as Sybil attack[31] and Eclipse attack[32]. Its workflow is as shown in Fig. 1.

- *Step 1*. A $enity_i$ participating in vehicular data sharing need to register the related information and apply for a DID to the *proxy*,then *proxy* creates a unique DID_i for the $enity_i$. Here we choose the DID generation method by multiple hashes and use base58 function to unify the length of the DID sequence.
- *Step 2*. The *issuer*(sk_s, pk_s) issues a credential based on the created DID_i and generates the $Signature_i(DID_i)$:

  $$Signature_i(DID_i) = Signature(sk_s, DID_i)$$

- *Step 3*. The system will upload the DID credential issued by the issuer to the blockchain. Only DID-related data is stored on the blockchain without storing private data.
- *Step 4*. When the $enity_i$ requests a certain IoV service, the service provider acting as a *verifier* need to verify the $enity_i$'s identity. The *verifier* can verify the identity by verifying whether the hash of the data applicant's DID is consistent with that in the obtained digital credential.
- *Step 5*. Given the announced DID'_i and the $enity_i$'s authorization,the *verifier* decryption $Signature_i(DID_i)$, and check if:

  $$\text{Hash}(DID'_i) == \text{Hash}(DID_i)$$

- *Step 6*. The *proxy* is also responsible for DID lifecycle management, such as revoking dishonest vehicle data sharing entities, etc.

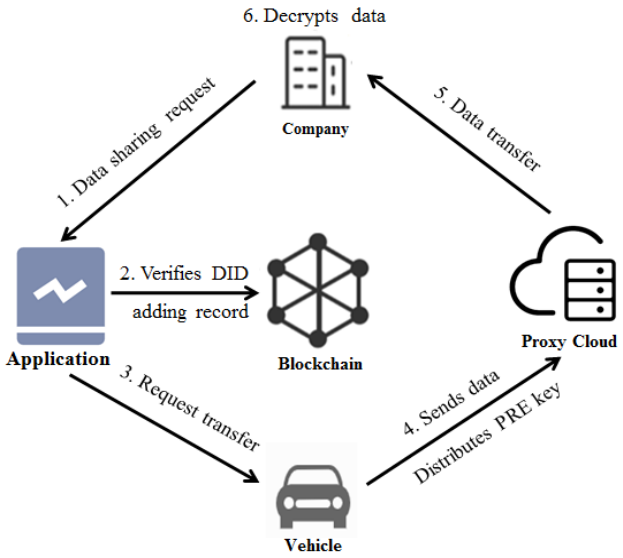

Fig. 2. Data sharing across entities

B. *Data sharing across entities*

After the process of DID setup in section A, an IoV provider such as a vehicle insurance service company can apply for vehicular data sharing in the proposed scheme. The workflow is mainly as shown in Fig. 2:

- *Step 1*: If a *company*(sk_A, pk_A) wants to view a target *vehicle*(sk_V, pk_V) 's data, it needs to send a data sharing request to the proposed platform application as the role of the system administrator.

  $$\text{KeyGen}(): a \in Z_q, (sk,pk) = (a,g^a)$$

- *Step 2*: After verifying the DID of the *company* that requests data, the platform application adds the data sharing record to the blockchain for easy auditing and traceability.

- *Step 3*: The application forwards the data sharing request to the *vehicle*. The *vehicle* has the right to review data transactions and makes corresponding processing. If data sharing is agreed, the transaction will be executed normally.

- *Step 4:*The *vehicle* encrypts the requested data with sk_V, sends the ciphertext to the data storage location such as the public cloud platform, and distributes the re-encrypted key($rk_{V->C}$) to the proxy re-encryption cloud.

  $rk_{V->C} = \text{ReKeyGen}(sk_V, pk_{V,N,t})$, (N: a number of fragments, t: a threshold,compute N fragments of the reencryption key,each of them calls KFrag)

- *Step 5*:After obtaining the *vehicle*'s authorization to share specified data and the re-encrypted key, the cloud proxy will query the data and re-encrypt the encrypted data from the *vehicle* with $rk_{V->C}$. Then the proxy will transmit the data to the *company* applying for data sharing.

- Step 6:Finally, the *company* decrypts the data owner's data with sk_A.

V. EXPERIMENTAL DESIGN

A. *Experimental setup*

We deployed a 4-nodes consortium blockchain named Fisco Bcos[33] on a server(RAM>256G, Storage capacity>30T) to simulate the task of accessing cross-entities data in the proposed system. This consortium blockchain provides a console for developers, which can directly initiate data sharing transactions in nodes.

Given the above-built blockchain, we set up the DID as described in Table 1 and tested the process of DID application. Table 1 shows the data structure of DID in the vehicular data sharing system. It includes the name of the data sharing participant, DID sequence, DID hash, user's type, credential serial number, issuance, and expiration date, and DID proxy, credential issuance agency.

As for proxy re-encryption, we used pyUmbral[34],the reference implementation of the Umbral, which is a proxy re-encryption network to empower privacy in decentralized systems. To generate an asymmetric key pair, we set SECP256K1 as a default curve to perform any ECC operations. In this experiment, we set different fragments and observe the performance of proxy re-encryption by controlling the size of the file. We issued a vehicle's data sharing task between nodes in the proposed system.

TABLE I
DID DATA STRUCTURES SAMPLE

FIELD	Description
Name	Vehicle AAA
DID	123456789abcdefghi#keys-1
Hash	86de8973acfe17d2f835956cef92624
Type	Vehicle/Data consumer
Credential serial	73ce2b24c5b0c60d3fb-01
Issue Date	Year/Month/Day
expire Date	Year/Month/Day
Proxy	XXX Proxy
Issuer	XXX administration Department

B. *Experimental result*

The file storage of this DID is only a few hundred bytes, and the time consumption for creating and indexing is in the level of milliseconds. The Fig. 3 shows the time required for DID related affairs such as DID creation, issuance, verification, etc. Overall, this process meets the speed requirements of the actual vehicular data sharing business.

In Fig. 4, we measured the time consumption of data transfer in four nodes' data sharing system using proxy re-encryption for different file sizes after the user authorization. At the same time, we also measured the average time consumption of file transfer in the case of different fragments. When the number of slices is fixed, with the gradual increase of shared data files, the time consumed also gradually increases, but on the whole, users still accept this delay time scale, it does not affect the availability of the system. In actual data sharing, if a large file is divided into multiple data pieces for transmission, its rate will gradually increase.

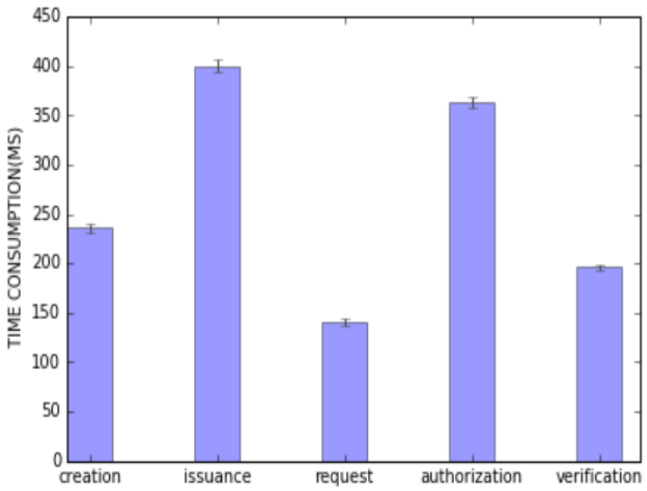

Fig. 3. Time consumption of DID setup in the proposed system based on the consortium blockchain. Normally, the DID setup is completed before the data sharing, which does not affect the data sharing process

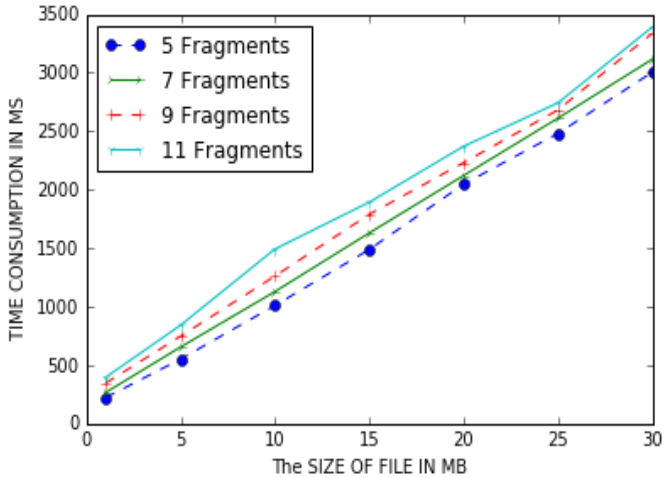

Fig. 4. Time consumption of sharing files based on proxy re-encrypting in the proposed system based on the consortium blockchain

VI. Security Analysis

In this part, we analyze the performance of the proposed scheme in terms of the basic performance of blockchain, privacy protection, and efficiency.

A. *Tamper-proof and non-repudiation*

The scheme is based on the blockchain technology, so all information in it is non-tamperable because all nodes of the system jointly maintain the same public ledger. The blockchain stores the overall hash value of vehicles' data, vehicles' data authorization credentials, etc., which can be traced and reviewed to facilitate vehicles owners' rights protection and return the rights of controlling data to vehicles owners.

As a trusted identity mechanism, DID is designed in the proposed scheme to achieve access to trusted entities. Each entity participating in data sharing has a unique identifier on the blockchain to avoid existing security attacks to a certain extent.

B. *Privacy protection*

- Anonymity: As with most blockchain systems, the proposed scheme will have an anonymous address for each vehicle, which is associated with the vehicle's key. All data sharing transactions in the scheme are anonymous and can greatly protect the privacy of vehicles owners.
- Privacy protection of sharing data: When accessing vehicle data across vehicles, it is a must to request vehicles' authorization. Proxy re-encryption is used to ensure data transmission security, which is compatible with existing IoV solutions using cloud service storage.

C. *Efficiency*

The scheme is based on the consortium blockchain. The transaction speed is much faster than the traditional public blockchain. In the IoV scenario, nodes can be dynamically joined or verified, and the size of a certain area is not particularly large. Secondly, data sharing is dynamic or time-sensitive, which is different from the traditional network architecture and causes less network overhead. And redundancy is not particularly large, the system administrator can choose to eliminate some malicious nodes or nodes with invalid identity, which is conducive to the construction of the entire data sharing ecology.

Secondly, we decouple the use and storage of data and put the process of data sharing off the blockchain. There are only authentication and records on the blockchain, which has greatly improved the efficiency and speed of vehicular data sharing. Besides, the overhead of proxy re-encryption meets the business needs of the actual data sharing scenes.

VII. Conclusion

In this paper, we propose a user-centric vehicular data sharing scheme based on consortium blockchain and proxy re-encryption. This scheme can realize data sharing across different IoV entities, and protect the data owners' privacy to some extent. It can help IoV service providers, vehicles, and other related organizations to achieve a balance between interests and data security, which is conducive to the development of IoV technology. In the end, we verified the proposed scheme through experiments and made a relevant experimental analysis. The most important thing is that the implementation of the scheme does great help to the existing vehicular data sharing problem solving and it can protect the privacy of users' data, which is also the original intention of the proposed scheme.

References

[1] D. Wang, D. Chen, B. Song, N. Guizani, X. Yu, X. Du, "From IoT to 5G I-IoT: The next generation IoT-based intelligent algorithms and 5G technologies", IEEE Communications Magazine., vol.56,no.10, pp.114 – 120, Oct. 2018.

[2] R. Z , L. W , L. H , et al. "A Novel Data Sharing Mechanism via Cloud-Based Dynamic Audit for Social Internet of Vehicles," International Conference on Internet of Vehicles, 2015

[3] J. Luo,W. Zhang, "The mp2p application in resource sharing policies of vanet," Wireless Internet Technology, 2015.

[4] S. Nakamoto,"Bitcoin: A Peer-to-Peer Electronic Cash System",2008

[5] P. Sharma, S. Yeon Moon, and J. Hyuk Park, "Block-VN: A Distributed Blockchain Based Vehicular Network Architecture in Smart City," JInf Process Syst., vol.13, no.1, p.184-195, 2017

[6] S. R. Mani Sekhar, G. M. Siddesh, S. Kalra, et al. "A Study of Use Cases for Smart Contracts Using Blockchain Technology," IJISSC, 2019

[7] V. Sharma, "An Energy-Efficient Transaction Model for the Blockchain-Enabled Internet of Vehicles (IoV)," in IEEE Communications Letters, v ol. 23, no. 2, pp. 246-249, Feb. 2019

[8] M. Lóo ,C. Mohamed ,K. Francine, "Towards a Blockchain-Based SD-Io V for Applications Authentication and Trust Management",2019.

[9] R. Sharma and S. Chakraborty, "BlockAPP: Using Blockchain for Authe ntication and Privacy Preservation in IoV," 2018 IEEE Globecom Works hops (GC Wkshps), Abu Dhabi, United Arab Emirates. , pp. 1-6,2018

[10] L. Mirador, H. Weiyan. "Implementing blockchain technology in the Int ernet of Vehicle (IoV)." ICEA,2019

[11] M. Onik, C. Kim, N. Lee, et al. "Privacy-aware blockchain for personal data sharing and tracking," Open Computer Science 9, pp. 80-91,2019.

[12] M. Shabani, "Blockchain-based platforms for genomic data sharing: a de-centralized approach in response to the governance problems," JAMIA 26(1),pp.76-80,2019.

[13] I. Makhdoom, I. Zhou, M. Abolhasan, et al. "Privacy Sharing: A Blockchain-based Framework for Integrity and Privacy-preserving Data Sharing in Smart Cities," ICETE (2),pp.363-371,2019.

[14] S. Wang, Y. Zhang, Y. Zhang, "A Blockchain-Based Framework for Data Sharing With Fine-Grained Access Control in Decentralized Storage Systems" IEEE Access 6, 2018.

[15] B. Zheng, L. Zhu, "Scalable and Privacy-Preserving Data Sharing Based on Blockchain," J. Comput. Sci. Technol. 33, pp. 557-567,2018.

[16] A. Zhang, X. Lin, "Towards Secure and Privacy-Preserving Data Sharing in e-Health Systems via Consortium Blockchain" J. Medical Systems 42,2018.

[17] X. Zhang, X. Chen, "Data Security Sharing and Storage Based on a Consortium Blockchain in a Vehicular Ad-hoc Network" IEEE Access 7,pp. 58241-58254 ,2019.

[18] M. Blaze, G. Bleumer, M. Strauss, "Divertible protocols and atomic proxy cryptography" Proc. Int. Conf. Theory Appl Cryptograph. Techn, pp. 127-144, 1998.

[19] N. Fotiou, I. Pittaras, V. Siris, George C,"Enabling Opportunistic Users in Multi-Tenant IoT Systems using Decentralized Identifiers and Permissioned Blockchains." International ACM Workshop on Security and Privacy for the Internet-of-Things., 2019

[20] K. Q , Lu R et al. "A privacy-preserving sensory data sharing scheme in Internet of Vehicles," Future generation computer systems.,2019.

[21] G. Sahil , K. Kuljeet ,K. Georges,G. Fran ̧ois , K. Neeraj ,H. Zhu. "Sec-IoV: A Multi-Stage Anomaly Detection Scheme for Internet of Vehicles",2019

[22] K. Fan , W. Wang,W. Jiang , et al. "Secure ultra-lightweight RFID mutual authentication protocol based on transparent computing for IoV," Peer to Peer Networking Applications, vol. 11,no. 4,pp.723-734, 2018,

[23] W. Chen, Y. Chen, X. Chen and Z. Zheng, "Toward Secure Data Sharing for the IoV: A Quality-Driven Incentive Mechanism With On-Chain and Off-Chain Guarantees," in IEEE Internet of Things Journal, vol. 7, no. 3, pp. 1625-1640, March 2020.

[24] Z. Tian, X. Gao, S. Su and J. Qiu, "Vcash: A Novel Reputation Framework for Identifying Denial of Traffic Service in Internet of Connected Vehicles," IEEE Internet of Things Journal., vol. 7, no. 5, pp. 3901-3909. May 2020.

[25] Z. Su, Y. Wang, Q. Xu and N. Zhang, "LVBS: Lightweight Vehicular Blockchain for Secure Data Sharing in Disaster Rescue," in IEEE Transactions on Dependable and Secure Computing, March. 2020

[26] R. Sharma and S. Chakraborty, "BlockAPP: Using Blockchain for Authentication and Privacy Preservation in IoV," IEEE Globecom Workshops (GC Wkshps), Abu Dhabi, United Arab Emirates, pp. 1-6, 2018, doi: 10.1109/GLOCOMW.2018.8644389.

[27] M. Egorov, M. Wilkison,"NuCypher KMS:Decentralized key management system,",2017

[28] J. Gu, B. Sun, X. Du, J. Wang, Y. Zhuang, Z. Wang, "Consortium Blockchain-based Malware Detection in Mobile Devices," IEEE Access, vol.: 6, no. 1, pp. 12118-12128, Feb. 2018

[29] H. Koshutanski, M. Ion, L. Telesca, "Distributed Identity Management Model for Digital Ecosystems" SECURWARE,pp.132-138,2007

[30] Q. Wang, W. Li, Z. Qin, "Proxy Re-Encryption in Access Control Framework of Information-Centric Networks," IEEE Access 7,pp.48417-48429,2019.

[31] S. Murali and A. Jamalipour, "A Lightweight Intrusion Detection for Sybil Attack Under Mobile RPL in the Internet of Things," in IEEE Internet of Things Journal, vol. 7, no. 1, pp. 379-388, Jan. 2020

[32] Q. Tan, Y. Gao, J. Shi, X. Wang, B. Fang and Z. Tian, Toward a Comprehensive Insight Into the Eclipse Attacks of Tor Hidden Services, IEEE Internet of Things Journal, vol. 6, no. 2, pp. 1584-1593, April 2019

[33] FISCO-BCOS [Online]. Available: https://github.com/FISCO-BCOS

[34] PyUmbral [Online]. Available: https://github.com/nucypher/pyUmbral

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